Module 03 | Beginner ML Project

End-to-End Tabular Machine Learning Baseline — 100 marks

Build one reproducible model workflow from the supplied synthetic student dataset. Predict **passed** (0 or 1) using features known before the final result. Use Python, Pandas, scikit-learn, Jupyter/Colab and a simple Streamlit prediction form.

Mandatory Submission Rules

- **Submit ONE ZIP file only.**
- **Rename the ZIP file StudentName_TrainingID.zip (example: Ali_12345.zip).**
- **Late submissions are not accepted.**

Required Steps and Deliverables

1. Inspect and choose features

Read student_learning.csv. Show rows, dtypes, target balance and missing values. Use passed as y. Exclude final_score because it reveals the result.

2. Split and baseline

Use a reproducible train/test split with stratification. Train DummyClassifier as a simple baseline.

3. Train at least two candidate models

Build preprocessing using ColumnTransformer + Pipeline. Train LogisticRegression and RandomForestClassifier or another justified classifier.

4. Evaluate fairly

Use the SAME held-out test set for the baseline and candidate models. Report accuracy, F1, confusion matrix and a brief error explanation.

5. Save and predict

Save the complete fitted pipeline with joblib. Reload it and predict new_student_records.csv without retraining.

6. Simple Streamlit interface

Provide an app where students can input feature values and see a predicted class. The interface may use the supplied starter template.

7. Document and submit

Include a runnable notebook, saved pipeline, evaluation summary, README, short model card, Streamlit script and dataset in one ZIP.

Student Marking Rubric — 100 Marks

Assessment criterion	Marks
Problem framing, features and leakage-aware split	20
Preprocessing pipeline and reproducibility	20
Baseline, two models, evaluation and error analysis	20
Streamlit prediction app / integration	15
Runnable code, files and documentation	10
Responsible-use notes, model card and demo	15
TOTAL	**100**

Important: This is synthetic classroom data. Predictions are for demonstration and must not be used to judge or penalize real students.